Catching the Habit in the Act: Protocol for a Pre-Registered N-of-1 Randomized Trial of Machine-Learning-Timed Nudges Against Mindless Scrolling

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Abstract

Most smartphone and social media use is started by the user, not by notifications, and people often describe stretches of it afterwards as meaningless. If mindless scrolling is largely a habit, a well-timed prompt that makes it visible might interrupt it. We present ScrollNudge, an open-source browser extension that measures scrolling behavior on social media sites in 10-second windows, collects in-the-moment self-reports of intentionality through experience sampling, trains a personal detector of mindless scrolling from those reports, and delivers a gentle, non-blocking nudge when the detector fires. We describe a pre-registered single-participant (n-of-1) randomized trial with a 7-day baseline followed by 14 days randomly allocated, 7 and 7, to nudge and no-nudge conditions. On no-nudge days the detector logs the moments at which it would have intervened ("shadow nudges"), which allows nudged moments to be compared with equivalent unnudged moments rather than with ordinary scrolling. Primary and secondary hypotheses concern stopping after a nudge, daily scrolling time and self-rated intentionality, and are tested with exact day-level randomization tests. A simulation-based power analysis shows that such a trial can detect only large effects: under our assumptions, 80% power requires roughly halving daily scrolling time or an odds ratio near 6 for stopping after a nudge. We therefore frame the trial as a pilot for estimating effect sizes, and we report the analysis plan, hypotheses and inference criteria before any outcome data are examined. Results will be added after data collection.

Keywords

digital wellbeing, nudges, habits, experience sampling, n-of-1 trial, just-in-time intervention, social media

1 Introduction

Concern about the time people spend on smartphones and social media is widespread, and it is often framed as a problem caused by devices and apps that interrupt and pull users in. Field evidence complicates this picture. In an in situ study using wearable cameras, Heitmayer and Lahlou found that 89% of smartphone interactions were initiated by the user rather than by a notification [4]. Earlier work showed that brief, repetitive "checking" sessions become habitual and act as gateways to longer use [16], and users describe much of their smartphone use as meaningless in retrospect, especially when it is habitual and passive [12].

If mindless scrolling is largely a habit, interventions should target the habit rather than the device. Habits are behaviors cued by context and performed with little conscious deliberation [19]. One way to interrupt such behavior is to make it visible at the moment it happens. Nudges, small changes in the choice environment that steer behavior without restricting options [18], are a natural fit: a short question such as "Still scrolling on purpose?" restores a moment of reflection without blocking anything.

Many digital self-control tools exist [13], but most act on fixed rules such as time limits, timers or blocking. Such rules ignore whether the user is scrolling purposefully at that moment, so they risk interrupting meaningful use and training users to dismiss them. Just-in-time adaptive interventions (JITAIs) instead aim to intervene when a person is both vulnerable and receptive [14]. Building one for scrolling requires a way to recognize mindless use from behavior alone, and an evaluation design that separates the effect of the intervention from the fact that it is delivered at unusual moments.

This paper describes ScrollNudge, a system and a pre-registered study protocol designed to meet both needs. Our contributions are:

- **An open-source system** that measures scrolling behavior in privacy-preserving 10-second windows, collects experience-sampling reports of intentionality, and delivers non-blocking nudges in the browser.
- **A personal detector of mindless scrolling** trained on the participant's own in-the-moment self-reports, evaluated with day-grouped cross-validation.
- **A randomized n-of-1 protocol with shadow nudges**, in which the detector keeps running on no-nudge days and logs the moments it would have intervened, giving a like-for-like comparison for the proximal effect of a nudge.
- **A simulation-based power analysis** that quantifies what a 14-day single-participant trial can and cannot detect, offering practical guidance for self-experimentation studies in HCI.

We publish the hypotheses, outcomes and analysis plan before any outcome data are analyzed. Results will be added to this paper after data collection is complete.

2 Background and Related Work

2.1 Habitual smartphone and social media use

Oulasvirta et al. described checking habits: brief, repeated sessions in which users quickly access dynamic content, which accounted for a large share of use and were associated with increased overall use [16]. Heitmayer and Lahlou combined wearable cameras with replay interviews, an approach known as subjective evidence-based ethnography [10], and found that most interactions were

self-initiated and that participants often regretted or could not explain them [4]. Lukoff et al. showed that use is experienced as less meaningful when it is habitual and passive [12], and that design features such as recommendations and autoplay reduce users' sense of agency on YouTube [11]. Together these findings suggest that the relevant target is the moment when use drifts from intentional to automatic.

2.2 Digital self-control tools and nudges

Lyngs et al. reviewed hundreds of digital self-control tools and organized them by the cognitive processes they target, noting that many rely on blocking or limits [13]. Hiniker et al. found that an app supporting goals and reflection reduced daily smartphone use [5]. Okeke et al. showed that a gentle, repeated vibration once a daily limit was exceeded reduced time on targeted apps [15], and Purohit et al. proposed digital nudges to make social media less addictive [17]. Kovacs et al. found that rotating interventions in the HabitLab browser extension increased effectiveness but also increased attrition [8], a reminder that interruptions have costs. Unlike limit-based nudges, ScrollNudge triggers on a learned, personal estimate of mindlessness rather than on elapsed time alone.

2.3 Just-in-time interventions and their evaluation

JITAIs adapt the type and timing of support to a person's changing state [14]. Evaluating them is difficult because interventions are delivered at moments selected by the system, which differ systematically from other moments. Micro-randomized trials address this by randomizing whether an intervention is delivered at each decision point [7]. Our shadow-nudge comparison follows the same logic at the level of days: the same decision rule runs in both conditions, and only the delivery of the message is randomized.

2.4 Self-experimentation in HCI

N-of-1 trials, in which a single person receives conditions in a randomized order, are an established way to learn what works for an individual [9]. HCI researchers have built systems that guide self-experiments for sleep [2] and food triggers [6]. Randomization tests are well suited to such designs because their validity follows from the randomization itself rather than from distributional assumptions [3]. Experience sampling, which collects brief self-reports at random moments in daily life, gives in-the-moment measures that avoid recall bias [1].

3 The ScrollNudge System

ScrollNudge is a Chrome extension (Manifest V3) with an analysis notebook (Figure 1). It runs on YouTube, Instagram, Facebook, X, Reddit, LinkedIn and TikTok, and does nothing on other sites.

3.1 Sensing

Every 10 seconds, a content script reports the distance scrolled (from wheel events and page scroll position), the number of scroll events and direction reversals, the number of clicks and key presses, and whether the tab was visible and focused. A background service worker groups windows into *sessions*: continuous activity on one site with no gap longer than 3 minutes. From the current session and its last 12 windows (2 minutes), it computes the features in Table 1.

Table 1: Detector features

Feature	Definition
elapsed_min	Minutes since the session started
speed_2m	Pixels scrolled per second over the last 2 minutes
clicks_pm	Clicks per minute over the last 2 minutes
reversals_pm	Scroll direction changes per minute
pause_frac	Share of the last 2 minutes with no scrolling
hour_sin, hour_cos	Time of day encoded on a circle
since_last_min	Minutes since the previous session on the site (capped at 240)

3.2 Experience sampling

After at least 2 minutes of activity, a check-in card asks: "How intentional is your scrolling right now?" on a 5-point scale from 1 (mindless) to 5 (on purpose), with a Skip option (Figure 2, left). Prompts are spaced at random intervals of 8 to 16 minutes of activity during baseline and 20 to 30 minutes during the experiment, to reduce burden once the detector is trained. Unanswered prompts close after 60 seconds and are logged as timeouts. Ratings of 1 or 2 are labeled *mindless*.

3.3 Detector

The detector is an L2-regularized logistic regression on standardized features. It is deliberately simple: it can be trained on a few dozen labels, its weights are interpretable, and it runs in a few lines of JavaScript inside the extension. Before training, a starter rule assigns a score of $\min(1, t/20)$ for session time t in minutes, halved when the click rate is 2 or more per minute, with a threshold of 0.75, which fires after roughly 15 minutes of low-interaction scrolling.

After baseline, the detector is trained on the baseline labels with balanced class weights. Performance is estimated with grouped cross-validation in which all labels from a day are held out together, so that the model is always tested on days it has not seen. The decision threshold is set to maximize F1 on the cross-validated predictions. Training is refused if there are fewer than 5 labels in either class or fewer than 3 days with labels, in which case the starter rule remains in use. The fitted means, standard deviations, coefficients and threshold are exported as `model.json` and loaded into the extension.

3.4 Nudge

A nudge is triggered when the tab is in use, the session has lasted at least 5 minutes, no nudge or shadow nudge has occurred in the past 10 minutes, and the detector score is at or above the threshold. The nudge reads "You've been on [site] for [N] minutes. Still scrolling on purpose?" with the options "Yes, continue" and "Take a break" (Figure 2, right). It never blocks the page. This design targets awareness rather than restriction, which is the mechanism under test: if the problem is habit rather than compulsion by the device, a reminder alone should help.

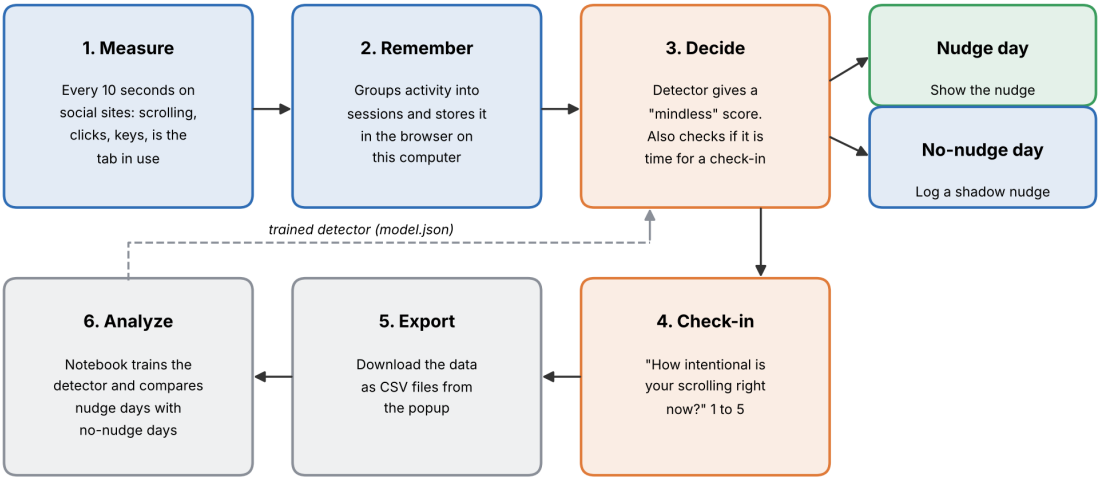


Figure 1: ScrollNudge pipeline. Activity is measured in 10-second windows, grouped into sessions and scored by the detector. On nudge days a high score shows a nudge; on other days it is logged as a shadow nudge. Experience-sampling check-ins provide labels for training the detector.

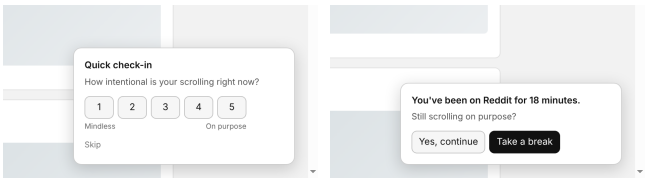


Figure 2: The experience-sampling check-in (left) and the nudge (right), shown in the corner of the page.

3.5 Privacy

No URLs, page content, typed text or screenshots are recorded; only counts and timings are stored, together with the site name. All data remain in the browser's local storage until the participant exports them. Exported data are excluded from the public repository, and only aggregate results will be shared.

4 Study Protocol

4.1 Design and participant

The study is a single-participant randomized trial with one participant, the author. The participant is a graduate student. Because the participant designed the system, he cannot be blind to its purpose; the design therefore relies on randomization and on comparing conditions, which are equally affected by this knowledge. Only the participant's own behavior is recorded. Before any results are published, the protocol will be submitted to the Rowan University Institutional Review Board for a determination.

4.2 Phases and randomization

The study lasts 21 days (Figure 3).

- **Baseline (days 1–7):** measurement and check-ins only. The starter rule runs, and any triggers are logged as shadow nudges.
- **Training (after day 7):** the detector is trained on baseline labels and loaded before or early on day 8.
- **Experiment (days 8–21):** 7 nudge days and 7 no-nudge days. The order is a uniformly random permutation drawn with a cryptographic random number generator when the extension is installed and stored before any outcome data exist.

Days are calendar days in local time. The participant is not shown the schedule in advance but learns the condition of a day once a nudge appears.

4.3 Hypotheses

H1 (primary, proximal). At detector-triggered moments, the probability of ending the session within 3 minutes is higher after a delivered nudge (nudge days) than after a shadow nudge (no-nudge days).

H2 (secondary, distal). Total scrolling time per day is lower on nudge days than on no-nudge days.

H3 (secondary). Mean self-rated intentionality is higher on nudge days than on no-nudge days.

Exploratory questions are: (E1) how well the detector predicts mindless reports on unseen days; (E2) whether intentionality declines with time in session, as the habit account predicts; and (E3) how the participant responds to nudges (continue, break or ignore).

4.4 Outcomes

- **Stopping within 3 minutes:** for each nudge or shadow nudge, whether the session's last active window occurs within 3 minutes of the trigger.

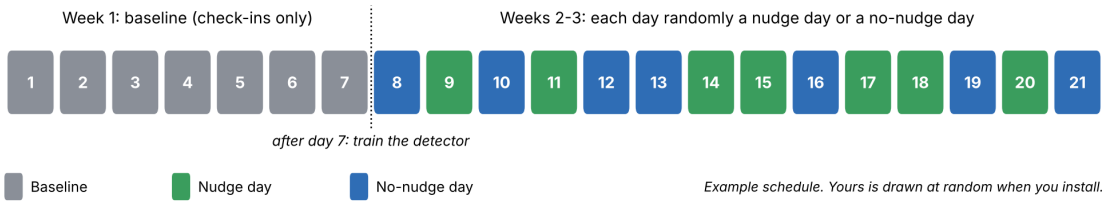


Figure 3: Study timeline. A 7-day baseline is followed by 14 experiment days, 7 nudge and 7 no-nudge, in an order drawn at random when the extension is installed. The schedule shown is an example.

- **Daily scrolling minutes:** the sum of window durations on tracked sites per day.
- **Intentionality:** the mean check-in rating per day.
- **Secondary descriptive measures:** sessions per day, mean session length, and sessions longer than 20 minutes.

4.5 Analysis plan

Because days, not moments, were randomized, all confirmatory tests are exact day-level randomization tests [3]. For each hypothesis the test statistic is computed for the observed allocation and for all $\binom{14}{7} = 3{,}432$ possible allocations of 7 nudge days among the 14 experiment days. The two-sided p-value is the share of allocations whose statistic is at least as extreme as the observed one. The statistics are:

- **H1:** the pooled stopping rate at nudge-day triggers minus the pooled rate at no-nudge-day triggers.
- **H2:** the mean daily minutes on nudge days minus the mean on no-nudge days.
- **H3:** the mean of daily mean ratings on nudge days minus the same on no-nudge days.

The significance level is $\alpha = 0.05$ for H1. H2 and H3 are tested at $\alpha = 0.05$ each and reported as secondary, without claiming confirmation from them alone. Given the small number of units, we will emphasize effect sizes: differences in rates, minutes and ratings with day-level bootstrap intervals, reported as descriptive.

Data handling. All 14 experiment days are analyzed. A day without tracked activity counts as 0 minutes for H2 and contributes no triggers to H1. Skipped and timed-out check-ins are excluded from rating analyses and reported as response rates. Triggers that occur when a check-in card is already open are not possible by design. Any deviation from this plan, including a late detector load or browser problems, will be reported.

4.6 Power analysis

To set expectations, we simulated the H1 and H2 tests (2,000 simulations each; code in the repository).

For H2, daily minutes were drawn from a log-normal distribution with a mean of 60 minutes and a day-to-day coefficient of variation (CV) of 0.4, 0.6 or 0.8, with nudge days reduced by 10% to 50%. For H1, the number of triggers per day was Poisson with a mean of 2, 4 or 6, the stopping probability without a nudge was 0.2, days varied with a random logit intercept (SD 0.5), and a nudge multiplied the odds of stopping by 1.5 to 6.

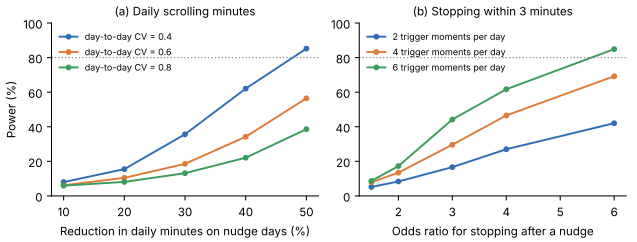


Figure 4: Simulated power of the day-level randomization tests at $\alpha = 0.05$. (a) Daily scrolling minutes by percentage reduction and day-to-day variation. (b) Stopping within 3 minutes by odds ratio and number of triggers per day. The dotted line marks 80% power.

Figure 4 shows the results. For H2, 80% power was reached only when daily minutes fell by 50% with low day-to-day variation (CV 0.4: 85.2% power); with CV 0.6, a 50% reduction gave about 56% power and a 30% reduction about 19%. For H1, power reached 84.9% only with 6 triggers per day and an odds ratio of 6; with 4 triggers per day, an odds ratio of 3 gave 29.6% power.

We also simulated a micro-randomized alternative in which each trigger on experiment days is randomized with probability 0.5 [7]. It increased power for H1 only modestly (for example, 35.6% instead of 29.6% with 4 triggers per day and an odds ratio of 3), and it cannot test the distal outcome H2. We therefore kept day-level randomization.

The trial is thus a pilot: it can detect large effects and will provide effect-size estimates and variance parameters for planning a larger study, but a non-significant result would not show that nudges are ineffective.

5 Results

Data collection began in November 2026 and is expected to end in early December 2026. Results will be reported here following the analysis plan above, including the detector's cross-validated performance, the confirmatory tests for H1 to H3, effect sizes, check-in response rates and any deviations from the protocol.

6 Discussion

6.1 Interpreting possible outcomes

The design separates two questions. A positive H1 with a null H2 would suggest that nudges interrupt individual episodes but that the time saved is made up later, a form of compensation. A positive H2 without H1 would suggest a cumulative effect of awareness rather than an in-the-moment one. Null results on both, given the limited power, would neither confirm nor rule out an effect, but they would bound plausible effect sizes. The exploratory question E2 tests a premise of the habit account directly: if intentionality ratings fall with time in session, the drift from purposeful to automatic use can be observed and timed.

6.2 Limitations

A single participant limits generalization; the study is intended to estimate effects for planning, not to establish population-level claims. Only laptop browsing is measured, while much social media use happens on phones. The participant is the system's designer and cannot be blinded, and awareness of the experiment may itself change behavior, although this affects both conditions. Check-ins are themselves brief interruptions that may act as nudges; they occur in both conditions and so do not bias the comparison, but they may make delivered nudges look less effective than they would be on their own. Finally, the detector is trained on a small number of self-reports and may be weak, in which case the intervention approaches a timer.

6.3 Future work

Natural extensions are a mobile version, a series of n-of-1 trials across many participants that can be combined to estimate both average and individual effects, comparisons of nudge wording and timing, and linking detector scores to replay interviews of wearable-camera footage [4, 10] to validate what the detector captures.

7 Conclusion

ScrollNudge combines in-the-moment self-reports, a personal detector of mindless scrolling and a randomized single-participant design with shadow nudges to test whether a gentle reminder can interrupt the habit of mindless scrolling. By publishing the protocol and analysis plan before examining outcomes, and by stating plainly what a 14-day trial can detect, we aim to make the eventual results, whatever they show, easy to interpret.

Open Science

The extension, analysis notebook and power-analysis code are available at https://github.com/aakashtiwarisolutions/scrollNudge. The study was pre-registered on the Open Science Framework on November 13, 2026, before data collection began: https://osf.io/hv8cg.

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